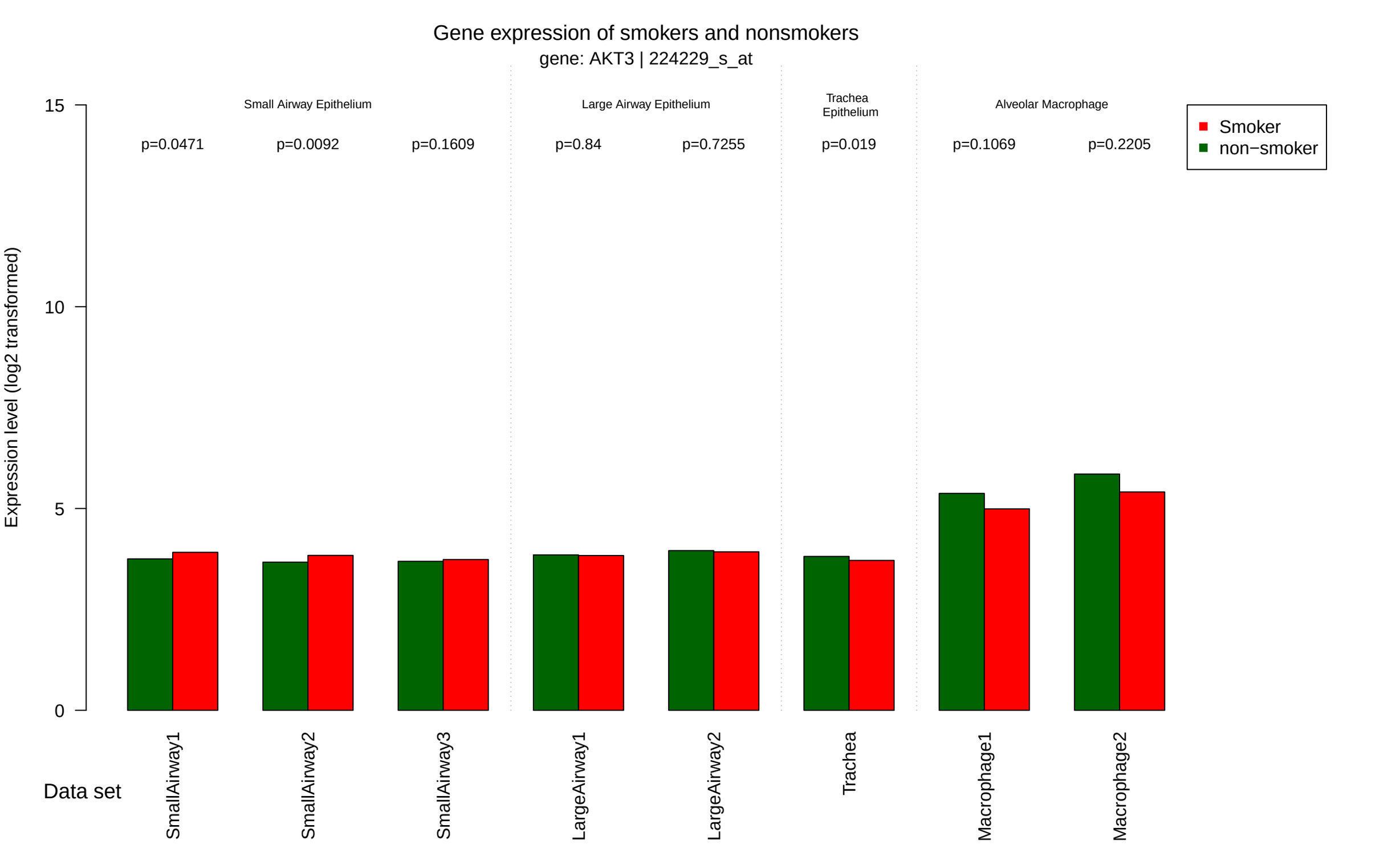
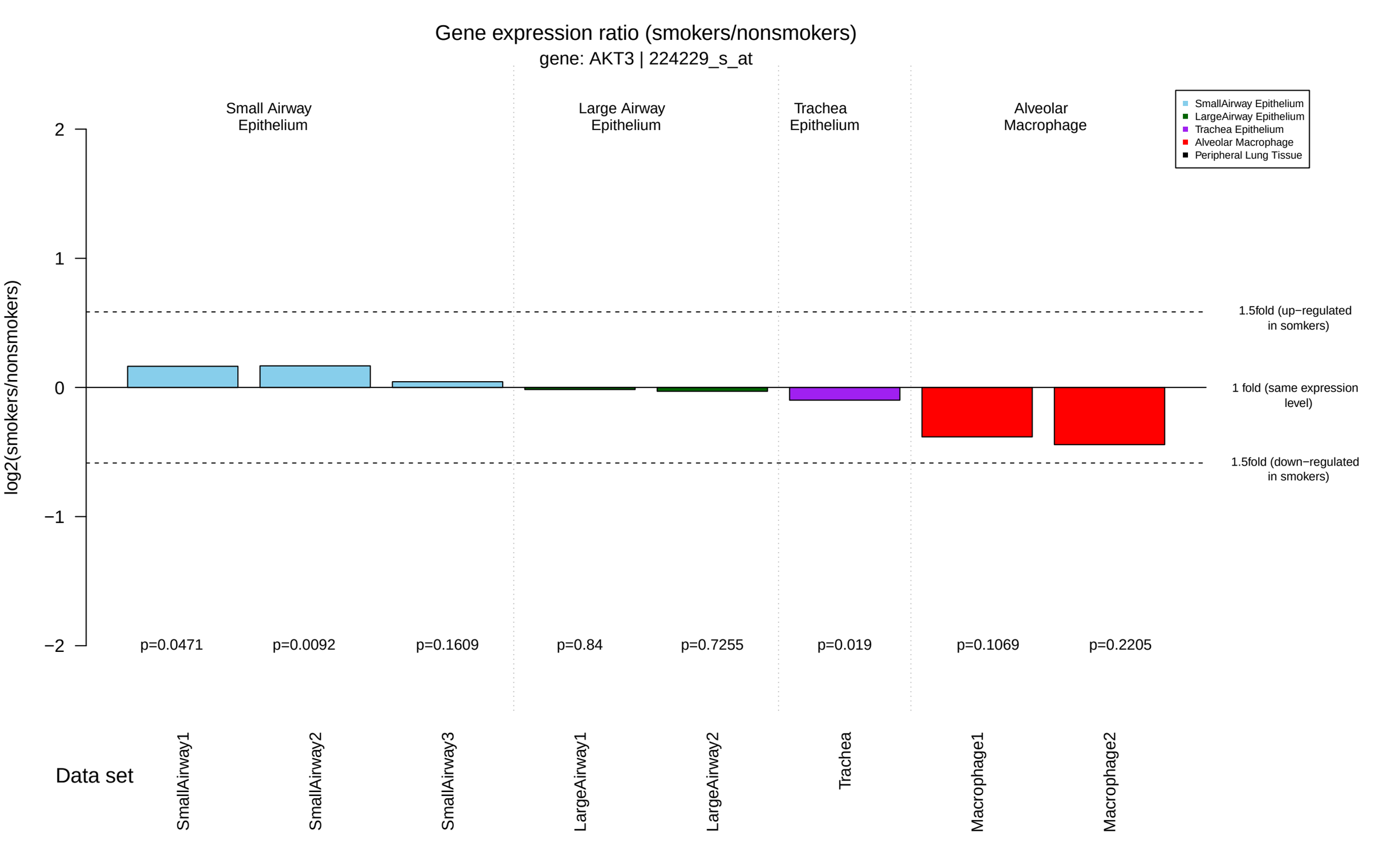
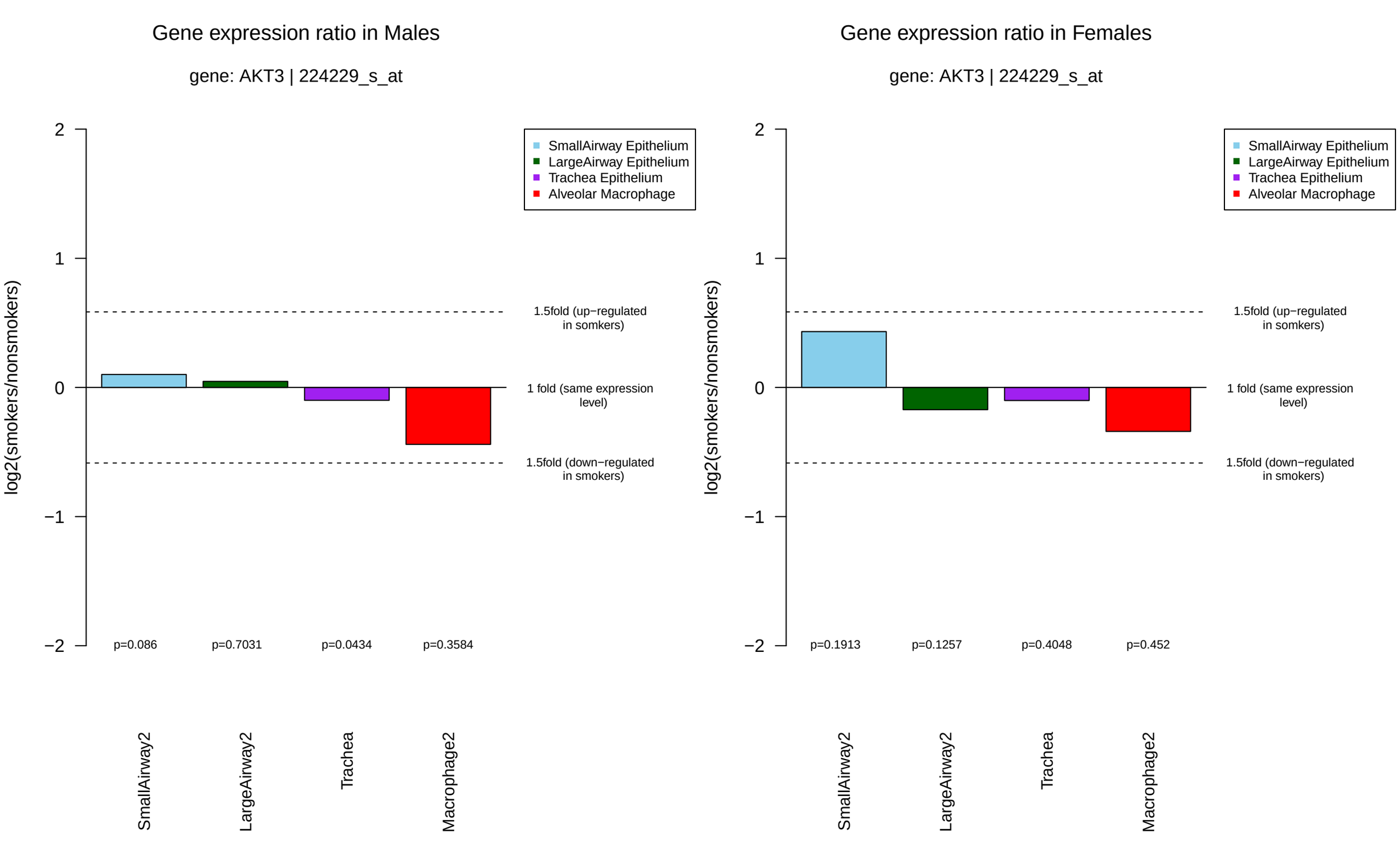
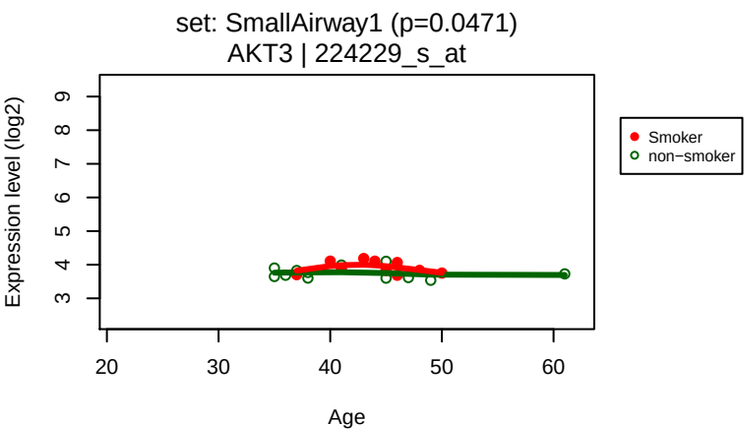
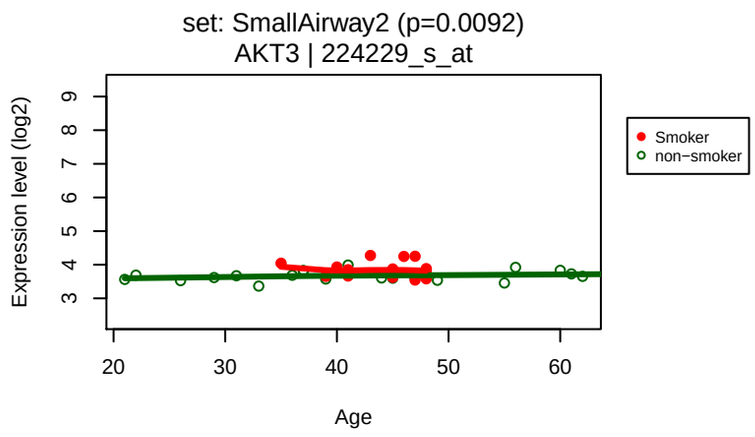
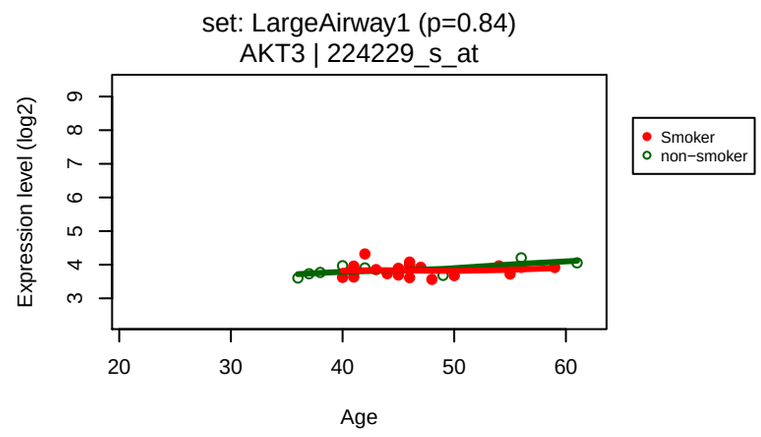

Gene expression of smokers and nonsmokers
gene: AKT3 | 224229_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0471
p=0.0092
p=0.1609
p=0.84
p=0.7255
p=0.019
p=0.1069
p=0.2205
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

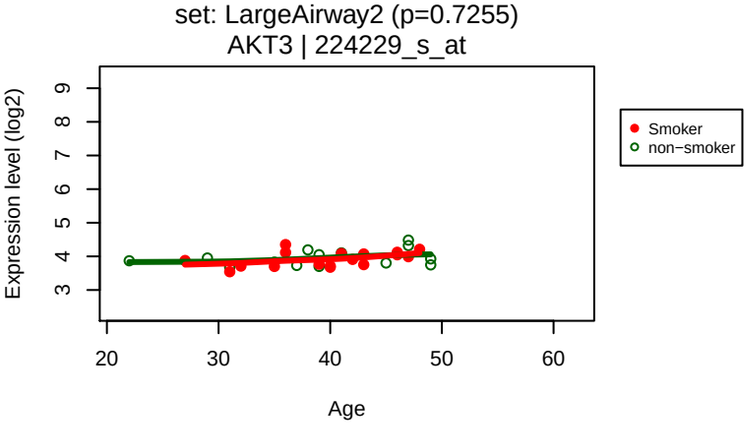

Gene expression ratio (smokers/nonsmokers)
gene: AKT3 | 224229_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0471
p=0.0092
p=0.1609
p=0.84
p=0.7255
p=0.019
p=0.1069
p=0.2205
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

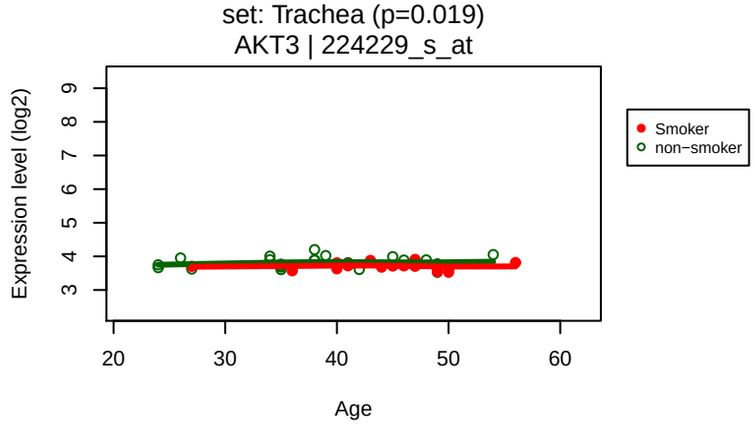

Gene expression ratio in Males
gene: AKT3 | 224229_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.086
p=0.7031
p=0.0434
p=0.3584
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKT3 | 224229_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1913
p=0.1257
p=0.4048
p=0.452
SmallAirway2
LargeAirway2
Trachea
Macrophage2

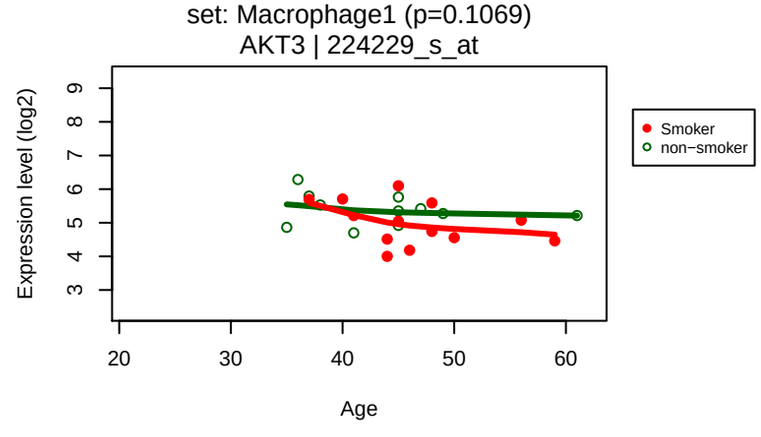

set: SmallAirway1 (p=0.0471)
AKT3 | 224229_s_at
Expression level (log2)
Age
Smoker
non−smoker

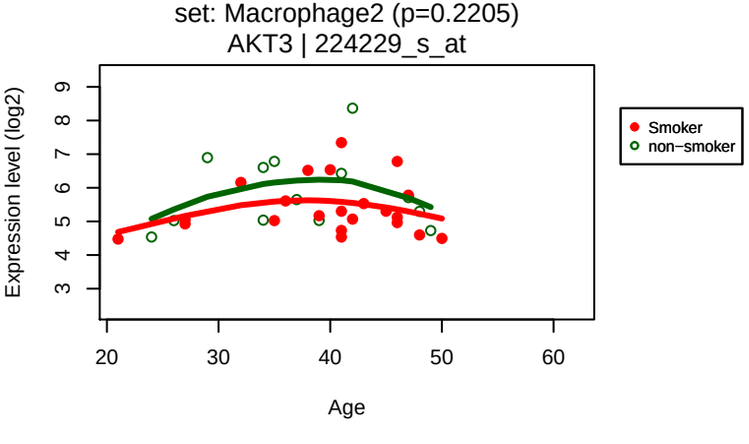

set: SmallAirway2 (p=0.0092)
AKT3 | 224229_s_at
Expression level (log2)
Age
Smoker
non−smoker

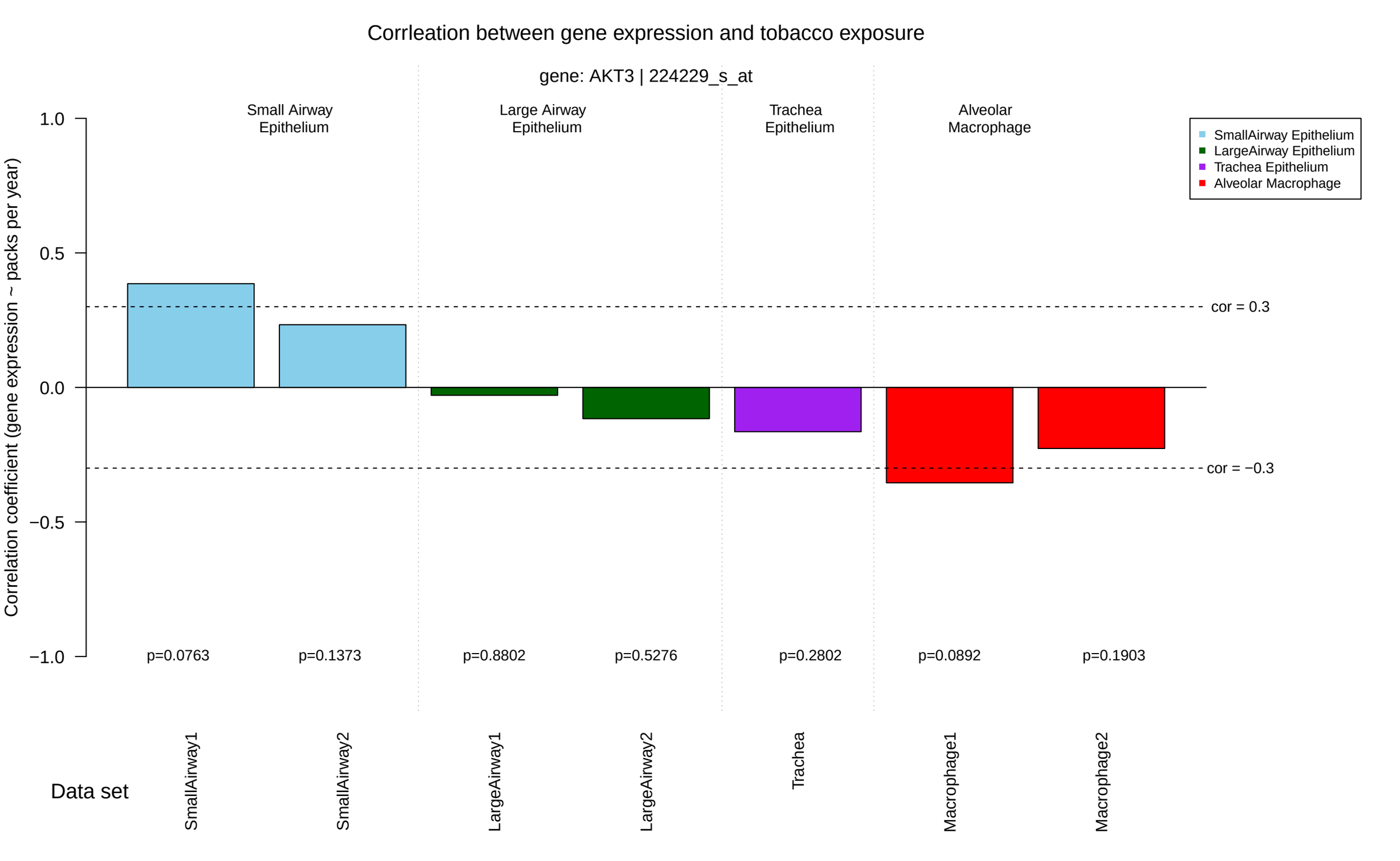

set: LargeAirway1 (p=0.84)
AKT3 | 224229_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7255)
AKT3 | 224229_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.019)
AKT3 | 224229_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1069)
AKT3 | 224229_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2205)
AKT3 | 224229_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKT3 | 224229_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0763
p=0.1373
p=0.8802
p=0.5276
p=0.2802
p=0.0892
p=0.1903
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2